

The Spanish Sun

With a name nearly synonymous with sun, Spain has become one of the European Union's top vacation destinations. The tourism industry has taken a liking to Spain's fair-weather resources, but with every passing sunny day, the country is missing out on millions of euros in renewable energy revenue. A wind industry leader, Spain has only 407 MW of installed PV production. This leaves this could-be sun factory of Europe around 2,200 MW behind Germany, a country less well known for sunbathing.

The Spanish government would like to "become a leader in the use of renewable energies," at least according to its president, José Luis Rodríguez Zapatero. His words are backed up by the renewable energy plan (PER in its Spanish notation), which calls for an already-exceeded 363 MW of installed solar PV capacity between 2005 and 2010, and a new building code (the Código Técnico de la Edificación [CTE]), that mandates solar hot water to be included in all new construction projects. Solar PV farms have become extremely popular, due in large part to gridconnected buyback rates of just over \$0.40/kWh for installations larger than 100 kW peak. A similar rate applies to small systems (less than 100 kW); however, bureaucratic delays and a complicated licensing process have prevented success on the scale of the German 100,000 Rooftops solar electricity program.

Ibiza, a small Mediterranean island with a big party reputation, is one of the sunniest places in Spain. Yet it finds itself in the shadows of the solar-energy movement. A recent report by the CNE (translated as the National Commission of Energy) rated the Balearic Islands, the chain of islands to which Ibiza belongs, as second to last in the country in renewable energy production. Grid-connected systems on the island would receive the same buyback rates as those on the mainland, but high

land prices and logistical difficulties have stifled such large-scale solar development.

That is not to say no one is installing renewable energy systems. Many of Europe's well-to-do dream of building a villa in this sun-drenched landscape. However, the same limited access to grid connections that may restrict the development of solar farms will also leave a countryside mansion in the dark. With price tags for access to high-tension wires reaching tens of thousands of euros, high-powered autonomous energy systems become an attractive option. A recently completed home, overlooking the island's 2,000-year-old salt-producing lakes, provides proof that living off-grid need not tarnish one's style.

The house is about 70 square meters (approximately 700 square feet) and has been outfitted to be as efficient as possible. Heating is radiant floor, the appliances are all low consumption, and the lighting scheme utilizes LED for indirect effects and low-consumption CFL lamps. The south-facing wall is made up of large windows, allowing heat and light to enter in the winter-time, but with an awning to provide protection from the summer sun.

Eric (full name omitted at his request), a long-standing member of the Ibiza live-music scene, wanted to build an off-grid home, but didn't

want to worry about watt-hours or butane refrigerators. He found his peace of mind through Erasmus Lob, a European solar technician, who has been designing renewable energy systems on Ibiza for the past six years. On an island known for its diversity, Erasmus has been able to provide power solutions utilizing renewable energy technology from all over the world. Eric's villa, equipped with solar modules from Japan; support structures, batteries, and a deep-well pump produced in Germany; and an inverter, solar charge regulator, and wind generators from the United States, is equipped to deliver these isolated sun worshipers with more than just a tan.

Water

Ibiza, especially in the summer months, has a dry climate. Land with a fresh water source is scarce, and if such land can be found, it is extremely



This Spanish home was traditionally built using modern technology.



A front porch with a view.

expensive. Most rural homes purchase their water by the truckload: 10,000 liters will cost around 85 euros. Being free from the electricity grid, Eric had no desire to dampen his home's self-sufficiency by relying on water deliveries. In order to achieve liquid autonomy, a 200-meter well was dug into the fresh-water table. Lob's job was not only to bring the precious fluid to the surface of the well, or even to the main house 10 meters uphill, but also to continue another 30 vertical meters (approximately 100 feet) to a 10,000-liter (2,640-gallon) storage tank.

Overcoming 240 meters (approximately 750 feet) of vertical head was no easy task, and for most renewable energy system designers would have necessitated the use of a generator. But where is the beauty in an off-grid home burning diesel in order to have water?

A Lorentz (HR-03H) helical rotor solar water pump allowed the generator to be used for backup purposes only. Helical rotor pumps are ideal for solar pumping applications in that they provide high lift capacity at low flow rates. Running directly off the solar panels, the pump controller (PS 1200) converts the DC supply into a three-phase AC current powering a brushless motor. The speed of the pump is regulated by its altering the frequency and voltage of the AC current in order to provide "low-gear" start-up (high current/low voltage) and "high-gear" (low current/high voltage) full-sun operation. Three Sanyo (Hit 210) 210W solar modules are connected in series, yielding an open-circuit voltage (VOC) of 150 VDC. During daylight hours the pump delivers a steady flow of 5–7 liters per minute. With Ibiza's five hours a day of year-round average sun, these figures will deliver an estimated 12,000 liters of solar-pumped water a week. With a price tag of 85 euros for 10,000 liters of water, this solar pump could lift just over 5,000 euros of water every year.

These figures certainly help to create a healthy return on the large investment required to dig a deep well; however, lowering the initial cost on the installation was also a high priority. Until recently, setting

a deep well pump was a labor-intensive process requiring the use of 6-meter-long iron pipes. Using a large machine, pipe after pipe would be lowered into the well and connected by rubber seals. This in effect created a long iron straw. Advances in material technology have allowed the iron pipes to be replaced by polymer-coated hosing. The installation process is as simple as lowering the fire hose-like flexible tubing into the well until the pump is submerged 20 meters (approximately 65 feet) below the water table. Such a system will cost less to maintain than a conventional installation, as corroded sections of the iron piping must be replaced every 5 or 6 years. The polymer-coated hosing is guaranteed for 10 years, and is predicted to last at least 20.

Wind

The heavy cement base securing the 10-ton water tank to the hillside was positioned slightly higher than necessary to provide water pressure of 3 bar (approximately 44 psi). This location minimized the number of trees that had to be felled in order to raise two Southwest Wind Power Air X, 400W, 48 VDC wind turbines. Here the mountain breeze blows with an average wind speed of 19 km/h (12 mph). The turbines selected have a rotor diameter of 1.17 meters and a start-up speed of 12 km/h (7.5 mph), ensuring their continuous function even in low wind conditions (which usually occur when it is sunny out). Each unit achieves a peak power output of 400 watts at wind speeds of 56 km/h (35 mph) and has an estimated monthly energy output of 30 kWh. The turbines have been installed using streetlight poles, which, being mass produced, cost less than specialized support structures and with the addition of tension cables, can withstand the lateral force required by the installation.

The wind turbines' microprocessor will continuously monitor the battery and is programmed to shut off the charge current automatically if the



The visual impact of the water tank is reduced with camouflage. The inset photo shows a closeup of one of two 400W wind turbines installed near the tank.

voltage of the system climbs to 57.8 VDC. Further mechanical protection will regulate the turbines' rpm; blade stall control is activated at wind speeds above 35 mph. Both features can be shut off with a manual stop switch located in the control room. The switch, by removing the turbine from the battery and then short-circuiting the wind generator, will allow the user to stall the unit for maintenance.

Solar PV

Three Sanyo (Hit 210) solar electric panels power the well pump, and eight generate most of the power for the house. Each Japanese-manufactured monocrystalline unit carries a rated efficiency of 18.9% and a VOC of 50 VDC. Four rows of two modules connected in series make up a 1.6-kW array with a VOC of 98 VDC. An American-made OutBack inverter/charger (FX3048E) with step-down capabilities connects the high-voltage



NIC BREACH

A control room provides easy access to renewable energy systems controls.

array to the low-voltage (48 VDC) battery. This technique saves on wire sizing and simplifies installation. The sophisticated maximum power point tracking (MPPT) charger, also made by OutBack (MX 60), will automatically record 64 days of system operation, which the user can access through a display located conveniently inside the home.

The rooftop installation uses an aluminum mount system from Germany that places the panels on their horizontal axis (thus decreasing visibility from ground level). In systems that are not connected to the grid, adjusting the angle of the solar module to conform to the angle of the sun's winter altitude stabilizes power

output during the course of the year. Fixed at a 41° angle, the solar array will maximize the system's winter output, when there is less sun. In a grid-connected system, the fixed mounting angle can be adjusted to maximize the system's controls annual output. In Ibiza, the optimum angle would be around 34°.

The brain of the system is the inverter/charger delivering 3 kW of continuous power and 6.7 kW peak. The inverter is programmed to automatically start an 8kW Pramac (8000) gasoline powered backup-up generator if the battery runs low, or if consumption spikes suddenly.

The battery pack consists of 24 Hawker (8 OPZS 800) deep-cycle cells with a rated capacity of 900 ampere-hours at a 100-hour discharge rate, and a total storage capacity of 38 kWh. The battery pack is sized to provide two full days of power for the house without receiving any charge.

Self-Sufficiency

With its own source of water, and electricity generated from wind and solar

power, this mountainside home has been designed to achieve a high level of self-sufficiency. A space has been left empty on the roof, however, where a solar-thermal collector will one day be installed. A gas system is currently in use, providing hot water on demand and supplying an in-floor heating system.

Like all large installations, a renewable energy system is a substantial investment. In Eric's case the total price tag came to just over 40,000 euros (about \$62,000 at the time of publication). In an environment such as Ibiza, however, where the local utility would have charged 130,000 euros to bring in the electricity lines, such systems are not only ecological, but economical as well. **H**_e

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